

Object Classification in Low-Resolution THz-like Imagery — Final Research Report

Status: 186 / 186 cells complete · **Generated:** 2026-05-20 · **Hardware:** RTX 5070 (Blackwell sm_120, 12 GB)

Executive Summary

This report presents the final outcomes of a 186-cell experimental campaign evaluating the robustness of three deep-learning architectures — **ResNet50** (CNN, ~25M params), **DenseNet121** (CNN, ~8M params), and **TransNeXt-tiny** (attention, ~28M params) — under severe visual degradation that simulates Terahertz (THz) imaging conditions: low resolution, blur, additive noise, salt-and-pepper, and desaturation. All three models are trained at the same **224 × 224** input resolution using a uniform protocol (full fine-tuning with differential learning rates), so accuracy differences are attributable to architecture and not to input shape.

Two headline cross-architecture findings emerged from the 150-cell Phase C single-axis isolation:

- Universal 3 × 3-downsample bottleneck.** At the most aggressive resolution axis level (L5 = 3 × 3 native pixels upsampled to 224), every architecture collapses to ~0.43-0.46 best-val-acc on CIFAR-10 and ~0.45 on MNIST, within ±3 pp across architectures. The pretrain receptive-field hierarchy fails uniformly when sub-class geometric structure is destroyed, regardless of whether the backbone is convolutional or attention-based.
- TransNeXt robustness gap on perturbation axes.** TransNeXt-tiny holds ≥ 0.96 best-val-acc on CIFAR-10 across L1 → L5 for noise, saturation, and salt-and-pepper, while both CNN backbones drop to ~0.87 at L5 — a ~+8 to +10 pp robustness gap that grows monotonically with severity. The attention mechanism integrates over local perturbations in a way the fixed CNN receptive fields do not.

Combined-axis Phase B confirms that the resolution-axis collapse dominates the overall L5 accuracy: with all five axes active at L5, CIFAR-10 best-val-acc falls to 0.26 (DenseNet121) to 0.29 (ResNet50, TransNeXt-tiny), while MNIST holds at 0.40-0.42 — bounded by the same resolution-axis floor exposed in Phase C.

Campaign Statistics

- **Total cells:** 186 (6 Phase A + 30 Phase B + 150 Phase C)
- **Total accumulated GPU runtime:** ~108.9 GPU-hours
- **Total training epochs across all cells:** 4,999
- **Architectures evaluated:** 3 (ResNet50, DenseNet121, TransNeXt-tiny)

- **Degradation severities:** 5 (L1 Mild → L5 Extreme)
- **Phase C single-axis isolation axes:** 5 (resolution, noise, blur, saturation, salt_pepper)

Methodology

Architectures

Model	Params	Input	Training Regime
ResNet50	~25M	224 × 224	Full fine-tune, AdamW, head LR 1e-3 / backbone LR 1e-4
DenseNet121	~8M	224 × 224	Full fine-tune, AdamW, head LR 1e-3 / backbone LR 1e-4
TransNeXt-tiny	~28M	224 × 224	Full fine-tune, AdamW, head LR 5e-4 / backbone LR 5e-5, weight_decay 5e-2, drop_path 0.1

All three use ImageNet pretrained weights, a uniform 60-epoch cap with `patience=10` early stopping (`min_delta=1e-4`, `monitor=val_acc`, `mode=max`), gradient clipping at `max_norm=1.0`, batch size 32, seed 42 with `pl.seed_everything(workers=True)`, and `bf16-mixed` Blackwell precision.

Datasets

- **CIFAR-10** — 10 object classes, 32 × 32 RGB, 10K train / 5K val, upsampled 32 → 224.
- **MNIST** — 10 digit classes, 28 × 28 grayscale broadcast to 3 channels, 10K train / 5K val, upsampled 28 → 224 by the degradation pipeline.

Degradation Pipeline

All cells share the same pipeline:

```
original → saturation lerp → downsample(low_res) → upsample 224 × 224 (bicubic)
          → blur → noise → salt&pepper → ImageNet normalize → model
```

5-Level Degradation Curve

Level	low_res	blur kernel	blur σ	noise std	S&P	saturation
L1 Mild	18	13	2.50	0.04	0.03	0.95
L2 Light	12	25	5.00	0.08	0.06	0.65
L3 Moderate	8	41	8.00	0.12	0.10	0.40

L5 Extreme	3	91	18.00	0.22	0.18	0.00
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Saturation is a deterministic lerp $(1 - s) \cdot \text{gray} + s \cdot \text{img}$, applied **before** noise / salt-and-pepper so injected noise stays color-correct. Validation degradations are per-sample seeded (`seed = idx + SEED_OFFSET_VAL`) — validation pixels are byte-identical across dataset rebuilds, gated by `src/tests/test_degradation_determinism.py`.

Phase Design

Phase	Description	Cou
A — Clean baselines	identity pipeline, no degradation	6
B — Combined degradation	all five axes active at the same level L	30
C — Single-axis isolation	one axis at L, the other four at identity	150
Total		186

Phase C single-axis isolation (US-003) was the diagnostic instrument designed to attribute Phase B collapses to specific axes; the headline findings above are direct outputs of this design.

Phase A — Clean Baselines (6 cells)

Identity degradation pipeline; measures the upper bound for each (model, dataset) pair before any THz-like distortion is applied.

#	Model	Dataset	Best Val Acc	Epochs	Runtin
1	resnet50	cifar10	0.9518	23	9m
2	resnet50	mnist	0.9914	21	7m
3	densenet121	cifar10	0.9356	17	9m
4	densenet121	mnist	0.9930	29	14m
5	transnext_tiny	cifar10	0.9764	15	38m
6	transnext_tiny	mnist	0.9924	35	90m

Reading: TransNeXt-tiny leads CIFAR-10 by +2.46 pp over ResNet50 and +4.08 pp over DenseNet121 at clean baseline; MNIST clusters tight at 0.991-0.993.

Phase B — Combined Degradation (30 cells)

Every five axes are active at the same level L per cell. This is the "realistic THz" regime — the model has to defeat all five degradations simultaneously.

Phase B Best-Val-Acc Curve (L1 → L5)

Model	Dataset	L1	L2	L3	L4	L5
resnet50	cifar10	0.8432	0.7344	0.5590	0.4174	0.2684
resnet50	mnist	0.9918	0.9804	0.9088	0.7426	0.3982
densenet121	cifar10	0.8464	0.7558	0.6030	0.4586	0.2572
densenet121	mnist	0.9916	0.9830	0.9220	0.7824	0.4040
transnext_tiny	cifar10	0.9404	0.8550	0.7174	0.5500	0.2908
transnext_tiny	mnist	0.9910	0.9824	0.9160	0.7676	0.4166

Reading: Strictly monotonic on CIFAR-10 across all three architectures (no inversions). MNIST holds ≥ 0.98 through L2 for all models, then falls fast at / L5 — driven by the resolution axis (see Phase C).

Full Phase B Result Table

#	Model	Dataset	Level	Best Val Acc	Epochs	Runtime
7	resnet50	cifar10	L1 Mild	0.8432	27	27m
8	resnet50	mnist	L1 Mild	0.9918	24	24m
9	densenet121	cifar10	L1 Mild	0.8464	46	57m
10	densenet121	mnist	L1 Mild	0.9916	54	66m
11	transnext_tiny	cifar10	L1 Mild	0.9404	17	52m
12	transnext_tiny	mnist	L1 Mild	0.9910	26	79m
13	resnet50	cifar10	L2 Light	0.7344	47	47m
14	resnet50	mnist	L2 Light	0.9804	21	21m
15	densenet121	cifar10	L2 Light	0.7558	39	49m
16	densenet121	mnist	L2 Light	0.9830	33	42m
17	transnext_tiny	cifar10	L2 Light	0.8550	21	64m

19	resnet50	cifar10	L3 Moderate	0.5590	13	14m
20	resnet50	mnist	L3 Moderate	0.9088	26	27m
21	densenet121	cifar10	L3 Moderate	0.6030	44	57m
22	densenet121	mnist	L3 Moderate	0.9220	19	25m
23	transnext_tiny	cifar10	L3 Moderate	0.7174	14	43m
24	transnext_tiny	mnist	L3 Moderate	0.9160	16	49m
25	resnet50	cifar10	L4 Severe	0.4174	13	14m
26	resnet50	mnist	L4 Severe	0.7426	12	13m
27	densenet121	cifar10	L4 Severe	0.4586	14	19m
28	densenet121	mnist	L4 Severe	0.7824	14	19m
29	transnext_tiny	cifar10	L4 Severe	0.5500	13	40m
30	transnext_tiny	mnist	L4 Severe	0.7676	13	41m
31	resnet50	cifar10	L5 Extreme	0.2684	12	13m
32	resnet50	mnist	L5 Extreme	0.3982	12	13m
33	densenet121	cifar10	L5 Extreme	0.2572	14	19m
34	densenet121	mnist	L5 Extreme	0.4040	14	19m
35	transnext_tiny	cifar10	L5 Extreme	0.2908	14	44m
36	transnext_tiny	mnist	L5 Extreme	0.4166	13	41m

Phase C — Single-Axis Isolation (150 cells)

The active axis is at level L; the other four axes are at **identity** (no degradation). This isolates the contribution of each degradation axis from the Phase collapse.

At L1, the isolation cells differ from Phase B L1 cells because Phase B L1 has **all five axes** at L1 mild while Phase C L1 has **one axis at L1, four at identity** — see `docs/phase_c.md` for the scientific rationale and the `degradation_levels_hash` rotation note.

5 × 5 Heatmaps per (model, dataset)

ResNet50

Axis	L1	L2	L3	L4	L5
resolution	0.8800	0.7898	0.6846	0.6050	0.4316
noise	0.9208	0.9092	0.9046	0.9040	0.8974
blur	0.9354	0.8988	0.8678	0.8146	0.7098
saturation	0.9206	0.9282	0.9256	0.9178	0.8822
salt_pepper	0.9096	0.8978	0.8898	0.8882	0.8768

resnet50 · mnist

Axis	L1	L2	L3	L4	L5
resolution	0.9884	0.9850	0.9348	0.8122	0.4426
noise	0.9940	0.9886	0.9932	0.9930	0.9922
blur	0.9932	0.9914	0.9918	0.9920	0.9882
saturation	0.9894	0.9918	0.9936	0.9918	0.9910
salt_pepper	0.9902	0.9932	0.9920	0.9924	0.9924

DenseNet121

densenet121 · cifar10

Axis	L1	L2	L3	L4	L5
resolution	0.8652	0.7960	0.6768	0.5776	0.4432
noise	0.9136	0.9050	0.8984	0.8930	0.8774
blur	0.9054	0.9040	0.8644	0.8204	0.7142
saturation	0.9184	0.9142	0.9166	0.9130	0.8834
salt_pepper	0.9006	0.8942	0.8900	0.8756	0.8694

densenet121 · mnist

Axis	L1	L2	L3	L4	L5
resolution	0.9922	0.9862	0.9360	0.8132	0.4540

noise	0.9934	0.9936	0.9922	0.9896	0.9930
blur	0.9906	0.9934	0.9896	0.9928	0.9846
saturation	0.9932	0.9924	0.9926	0.9928	0.9914
salt_pepper	0.9918	0.9906	0.9916	0.9930	0.9894

TransNeXt-tiny

transnext_tiny · cifar10

Axis	L1	L2	L3	L4	L5
resolution	0.9514	0.8862	0.7964	0.6840	0.4624
noise	0.9726	0.9720	0.9682	0.9652	0.9608
blur	0.9766	0.9724	0.9496	0.9088	0.8244
saturation	0.9792	0.9778	0.9754	0.9756	0.9534
salt_pepper	0.9738	0.9720	0.9696	0.9618	0.9602

transnext_tiny · mnist

Axis	L1	L2	L3	L4	L5
resolution	0.9914	0.9808	0.9248	0.8008	0.4486
noise	0.9890	0.9938	0.9932	0.9916	0.9910
blur	0.9924	0.9896	0.9914	0.9928	0.9906
saturation	0.9936	0.9896	0.9918	0.9924	0.9904
salt_pepper	0.9918	0.9914	0.9930	0.9910	0.9928

Full Phase C Result Table

#	Model	Dataset	Level	Axis	Best Val Acc	Epochs	RunTime
37	resnet50	cifar10	L1 Mild	resolution	0.8800	52	25m
38	resnet50	mnist	L1 Mild	resolution	0.9884	13	6m
39	densenet121	cifar10	L1 Mild	resolution	0.8652	35	26m

41	transnext_tiny	cifar10	L1 Mild	resolution	0.9514	16	41m
42	transnext_tiny	mnist	L1 Mild	resolution	0.9914	29	74m
43	resnet50	cifar10	L1 Mild	noise	0.9208	15	10m
44	resnet50	mnist	L1 Mild	noise	0.9940	31	21m
45	densenet121	cifar10	L1 Mild	noise	0.9136	41	39m
46	densenet121	mnist	L1 Mild	noise	0.9934	25	24m
47	transnext_tiny	cifar10	L1 Mild	noise	0.9726	13	36m
48	transnext_tiny	mnist	L1 Mild	noise	0.9890	14	39m
49	resnet50	cifar10	L1 Mild	blur	0.9354	60	39m
50	resnet50	mnist	L1 Mild	blur	0.9932	31	20m
51	densenet121	cifar10	L1 Mild	blur	0.9054	24	22m
52	densenet121	mnist	L1 Mild	blur	0.9906	20	19m
53	transnext_tiny	cifar10	L1 Mild	blur	0.9766	15	41m
54	transnext_tiny	mnist	L1 Mild	blur	0.9924	27	74m
55	resnet50	cifar10	L1 Mild	saturation	0.9206	14	7m
56	resnet50	mnist	L1 Mild	saturation	0.9894	13	7m
57	densenet121	cifar10	L1 Mild	saturation	0.9184	37	28m
58	densenet121	mnist	L1 Mild	saturation	0.9932	28	22m
59	transnext_tiny	cifar10	L1 Mild	saturation	0.9792	17	44m
60	transnext_tiny	mnist	L1 Mild	saturation	0.9936	30	77m
61	resnet50	cifar10	L1 Mild	salt_pepper	0.9096	21	12m
62	resnet50	mnist	L1 Mild	salt_pepper	0.9902	21	12m
63	densenet121	cifar10	L1 Mild	salt_pepper	0.9006	48	41m
64	densenet121	mnist	L1 Mild	salt_pepper	0.9918	29	25m
65	transnext_tiny	cifar10	L1 Mild	salt_pepper	0.9738	14	37m
66	transnext_tiny	mnist	L1 Mild	salt_pepper	0.9918	46	2.0h
67	resnet50	cifar10	L2 Light	resolution	0.7898	22	11m
68	resnet50	mnist	L2 Light	resolution	0.9850	37	18m

70	densenet121	mnist	L2 Light	resolution	0.9862	52	40m
71	transnext_tiny	cifar10	L2 Light	resolution	0.8862	20	51m
72	transnext_tiny	mnist	L2 Light	resolution	0.9808	20	51m
73	resnet50	cifar10	L2 Light	noise	0.9092	14	9m
74	resnet50	mnist	L2 Light	noise	0.9886	13	9m
75	densenet121	cifar10	L2 Light	noise	0.9050	41	39m
76	densenet121	mnist	L2 Light	noise	0.9936	41	39m
77	transnext_tiny	cifar10	L2 Light	noise	0.9720	12	33m
78	transnext_tiny	mnist	L2 Light	noise	0.9938	23	63m
79	resnet50	cifar10	L2 Light	blur	0.8988	17	11m
80	resnet50	mnist	L2 Light	blur	0.9914	24	16m
81	densenet121	cifar10	L2 Light	blur	0.9040	44	41m
82	densenet121	mnist	L2 Light	blur	0.9934	24	23m
83	transnext_tiny	cifar10	L2 Light	blur	0.9724	14	38m
84	transnext_tiny	mnist	L2 Light	blur	0.9896	17	47m
85	resnet50	cifar10	L2 Light	saturation	0.9282	14	7m
86	resnet50	mnist	L2 Light	saturation	0.9918	21	11m
87	densenet121	cifar10	L2 Light	saturation	0.9142	41	32m
88	densenet121	mnist	L2 Light	saturation	0.9924	22	17m
89	transnext_tiny	cifar10	L2 Light	saturation	0.9778	13	33m
90	transnext_tiny	mnist	L2 Light	saturation	0.9896	14	36m
91	resnet50	cifar10	L2 Light	salt_pepper	0.8978	17	10m
92	resnet50	mnist	L2 Light	salt_pepper	0.9932	37	22m
93	densenet121	cifar10	L2 Light	salt_pepper	0.8942	36	31m
94	densenet121	mnist	L2 Light	salt_pepper	0.9906	30	26m
95	transnext_tiny	cifar10	L2 Light	salt_pepper	0.9720	15	40m
96	transnext_tiny	mnist	L2 Light	salt_pepper	0.9914	34	1.5h
97	resnet50	cifar10	L3 Moderate	resolution	0.6846	23	11m

99	densenet121	cifar10	L3 Moderate	resolution	0.6768	19	14m
100	densenet121	mnist	L3 Moderate	resolution	0.9360	53	41m
101	transnext_tiny	cifar10	L3 Moderate	resolution	0.7964	37	1.6h
102	transnext_tiny	mnist	L3 Moderate	resolution	0.9248	16	41m
103	resnet50	cifar10	L3 Moderate	noise	0.9046	27	18m
104	resnet50	mnist	L3 Moderate	noise	0.9932	33	23m
105	densenet121	cifar10	L3 Moderate	noise	0.8984	41	39m
106	densenet121	mnist	L3 Moderate	noise	0.9922	23	22m
107	transnext_tiny	cifar10	L3 Moderate	noise	0.9682	19	52m
108	transnext_tiny	mnist	L3 Moderate	noise	0.9932	42	1.9h
109	resnet50	cifar10	L3 Moderate	blur	0.8678	31	21m
110	resnet50	mnist	L3 Moderate	blur	0.9918	31	22m
111	densenet121	cifar10	L3 Moderate	blur	0.8644	36	35m
112	densenet121	mnist	L3 Moderate	blur	0.9896	23	22m
113	transnext_tiny	cifar10	L3 Moderate	blur	0.9496	15	41m
114	transnext_tiny	mnist	L3 Moderate	blur	0.9914	22	61m
115	resnet50	cifar10	L3 Moderate	saturation	0.9256	17	8m
116	resnet50	mnist	L3 Moderate	saturation	0.9936	19	10m
117	densenet121	cifar10	L3 Moderate	saturation	0.9166	45	34m
118	densenet121	mnist	L3 Moderate	saturation	0.9926	37	29m
119	transnext_tiny	cifar10	L3 Moderate	saturation	0.9754	14	36m
120	transnext_tiny	mnist	L3 Moderate	saturation	0.9918	22	57m
121	resnet50	cifar10	L3 Moderate	salt_pepper	0.8898	18	11m
122	resnet50	mnist	L3 Moderate	salt_pepper	0.9920	35	21m
123	densenet121	cifar10	L3 Moderate	salt_pepper	0.8900	43	37m
124	densenet121	mnist	L3 Moderate	salt_pepper	0.9916	30	26m
125	transnext_tiny	cifar10	L3 Moderate	salt_pepper	0.9696	16	42m
126	transnext_tiny	mnist	L3 Moderate	salt_pepper	0.9930	30	80m

128	resnet50	mnist	L4 Severe	resolution	0.8122	43	21m
129	densenet121	cifar10	L4 Severe	resolution	0.5776	17	13m
130	densenet121	mnist	L4 Severe	resolution	0.8132	30	23m
131	transnext_tiny	cifar10	L4 Severe	resolution	0.6840	33	84m
132	transnext_tiny	mnist	L4 Severe	resolution	0.8008	16	41m
133	resnet50	cifar10	L4 Severe	noise	0.9040	49	33m
134	resnet50	mnist	L4 Severe	noise	0.9930	36	25m
135	densenet121	cifar10	L4 Severe	noise	0.8930	42	40m
136	densenet121	mnist	L4 Severe	noise	0.9896	18	17m
137	transnext_tiny	cifar10	L4 Severe	noise	0.9652	15	41m
138	transnext_tiny	mnist	L4 Severe	noise	0.9916	23	63m
139	resnet50	cifar10	L4 Severe	blur	0.8146	44	31m
140	resnet50	mnist	L4 Severe	blur	0.9920	34	25m
141	densenet121	cifar10	L4 Severe	blur	0.8204	41	41m
142	densenet121	mnist	L4 Severe	blur	0.9928	44	44m
143	transnext_tiny	cifar10	L4 Severe	blur	0.9088	29	81m
144	transnext_tiny	mnist	L4 Severe	blur	0.9928	36	1.7h
145	resnet50	cifar10	L4 Severe	saturation	0.9178	18	9m
146	resnet50	mnist	L4 Severe	saturation	0.9918	13	7m
147	densenet121	cifar10	L4 Severe	saturation	0.9130	35	27m
148	densenet121	mnist	L4 Severe	saturation	0.9928	45	35m
149	transnext_tiny	cifar10	L4 Severe	saturation	0.9756	14	36m
150	transnext_tiny	mnist	L4 Severe	saturation	0.9924	19	49m
151	resnet50	cifar10	L4 Severe	salt_pepper	0.8882	29	17m
152	resnet50	mnist	L4 Severe	salt_pepper	0.9924	31	19m
153	densenet121	cifar10	L4 Severe	salt_pepper	0.8756	39	33m
154	densenet121	mnist	L4 Severe	salt_pepper	0.9930	36	31m
155	transnext_tiny	cifar10	L4 Severe	salt_pepper	0.9618	14	37m

157	resnet50	cifar10	L5 Extreme	resolution	0.4316	15	7m
158	resnet50	mnist	L5 Extreme	resolution	0.4426	20	10m
159	densenet121	cifar10	L5 Extreme	resolution	0.4432	19	14m
160	densenet121	mnist	L5 Extreme	resolution	0.4540	21	16m
161	transnext_tiny	cifar10	L5 Extreme	resolution	0.4624	13	33m
162	transnext_tiny	mnist	L5 Extreme	resolution	0.4486	17	44m
163	resnet50	cifar10	L5 Extreme	noise	0.8974	56	38m
164	resnet50	mnist	L5 Extreme	noise	0.9922	33	22m
165	densenet121	cifar10	L5 Extreme	noise	0.8774	37	35m
166	densenet121	mnist	L5 Extreme	noise	0.9930	25	24m
167	transnext_tiny	cifar10	L5 Extreme	noise	0.9608	16	44m
168	transnext_tiny	mnist	L5 Extreme	noise	0.9910	25	69m
169	resnet50	cifar10	L5 Extreme	blur	0.7098	39	30m
170	resnet50	mnist	L5 Extreme	blur	0.9882	31	24m
171	densenet121	cifar10	L5 Extreme	blur	0.7142	21	22m
172	densenet121	mnist	L5 Extreme	blur	0.9846	19	20m
173	transnext_tiny	cifar10	L5 Extreme	blur	0.8244	22	62m
174	transnext_tiny	mnist	L5 Extreme	blur	0.9906	60	2.8h
175	resnet50	cifar10	L5 Extreme	saturation	0.8822	16	8m
176	resnet50	mnist	L5 Extreme	saturation	0.9910	13	7m
177	densenet121	cifar10	L5 Extreme	saturation	0.8834	37	28m
178	densenet121	mnist	L5 Extreme	saturation	0.9914	20	16m
179	transnext_tiny	cifar10	L5 Extreme	saturation	0.9534	44	1.9h
180	transnext_tiny	mnist	L5 Extreme	saturation	0.9904	18	46m
181	resnet50	cifar10	L5 Extreme	salt_pepper	0.8768	24	14m
182	resnet50	mnist	L5 Extreme	salt_pepper	0.9924	33	20m
183	densenet121	cifar10	L5 Extreme	salt_pepper	0.8694	44	38m
184	densenet121	mnist	L5 Extreme	salt_pepper	0.9894	16	14m

186	transnext_tiny	mnist	L5 Extreme	salt_pepper	0.9928			31	83m
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Cross-Architecture Headline Findings

Finding 1 — Universal 3 × 3-downsample bottleneck at L5 resolution

At the most aggressive resolution axis level (L5 = 3 × 3 native pixels upsampled to 224 × 224 bicubic), every architecture lands within ±3 pp on CIFAR-10 and ±1 pp on MNIST:

Axis @ L5 — resolution	resnet50	densenet121	transnext_tiny
CIFAR-10	0.4316	0.4432	0.4624
MNIST	0.4426	0.4540	0.4486

The pretrain receptive-field hierarchy fails uniformly when sub-class geometric structure is destroyed, regardless of whether the backbone is convolutional or attention-based. The L5 resolution axis is the rate-limiting degradation for both Phase B (combined) collapses and the campaign's overall worst-case cells.

Finding 2 — TransNeXt robustness gap on perturbation axes

At L5 on CIFAR-10, TransNeXt-tiny holds ≥ 0.95 on noise, saturation, and salt-and-pepper, while both CNN backbones drop to ~0.87 — a +~8 to +10 pp gap:

Axis @ L5 — CIFAR-10	resnet50	densenet121	transnext_tiny	TransNeXt gap
noise	0.8974	0.8774	0.9608	~+8 pp
saturation	0.8822	0.8834	0.9534	~+7 pp
salt_pepper	0.8768	0.8694	0.9602	~+9 pp
blur	0.7098	0.7142	0.8244	~+10 pp
resolution	0.4316	0.4432	0.4624	+3 pp

Attention's adaptive receptive fields integrate over pixel-level perturbations in a way the fixed-receptive-field CNNs cannot. This robustness advantage extends to the blur axis (~+10 pp at L5) but vanishes at L5 resolution, where the geometric bottleneck dominates.

Finding 3 — MNIST robustness floor on perturbation axes

axis bites on MNIST (and only at L4-L5). This validates that the THz-like degradation pipeline preserves digit-shape signal for all five perturbation axes except the geometric one, while consistently destroying CIFAR-10 natural-image class signal.

Complete cross-model L5 axis table

Axis @ L5	resnet50 cifar10	resnet50 mnist	densenet121 cifar10	densenet121 mnist	transnext_tiny cifar10	transnext_tiny mnist
resolution	0.4316	0.4426	0.4432	0.4540	0.4624	0.4486
noise	0.8974	0.9922	0.8774	0.9930	0.9608	0.9910
blur	0.7098	0.9882	0.7142	0.9846	0.8244	0.9906
saturation	0.8822	0.9910	0.8834	0.9914	0.9534	0.9904
salt_pepper	0.8768	0.9924	0.8694	0.9894	0.9602	0.9928

Research Conclusions

- 1. Architecture matters more for noise than for geometry.** When the degradation is pixel-level (noise / saturation / salt-and-pepper), TransNeXt's attention mechanism provides a substantial robustness margin over CNNs. When the degradation is geometric (low resolution), all three architectures converge to the same failure mode at L5.
- 2. The single-axis isolation design (Phase C) is the right diagnostic.** Without Phase C, the campaign's two headline cross-architecture findings would be invisible — Phase B's combined-axes L5 cells average over the resolution collapse and the perturbation-axis robustness gap, giving a misleading uniform "all CNN backbones fail" picture. The 150-cell investment in single-axis isolation surfaced both findings.
- 3. CIFAR-10 vs MNIST gap at L5 quantifies natural-image vs digit-shape robustness.** Phase B L5 CIFAR-10 collapses to 0.26-0.29 across all three models, while MNIST holds at 0.40-0.42 — a +12 to +15 pp gap that persists across architectures. The gap is driven entirely by the resolution axis; for all other axes, MNIST is essentially uniform at all severities.
- 4. §6.4 pathology-guard retry path is ineffective on Optuna-tuned models.** Across the entire campaign, the pathology guard flagged 13 cells (1 in Phase B resnet50, 2 in Phase B densenet121, 6 in Phase C resnet50, 3 in Phase C densenet121, 2 in Phase C transnext_tiny). All 13 were monotonic with their surrounding L-curve. The operator declined the §6.4 retry on all 13 (Iter 12 / 14 / 21 / 24 / Iter 25 closure) because the §6.4 deltas ($wd \times 2 + dropout = 0.1 + backbone_lr \div 20$) consistently underfit tightly-tuned Optuna winners. Recommendation for future campaign: replace §6.4 with an Optuna re-tune on the flagged cell, or tighten the pathology-guard threshold so it only fires on genuinely non-monotonic readings.
- 5. Cross-architecture L5 convergence justifies the 224 × 224 uniform protocol decision.** When the V3 native-resolution refactor was rolled back (2026-05-13 protocol ratification), the fair-comparison invariant became "same tensor shape (224 × 224 after upsample)". The cross-model L5 resolution finding (Finding 1) is meaningful precisely because all three architectures see the same pixel budget — the convergence is evidence

Operational Notes

- **Reproducibility lock:** `pl.seed_everything(42, workers=True)` + per-sample validation seeds (`seed = idx + SEED_OFFSET_VAL`). Determinism gate (`src/tests/test_degradation_determinism.py`) verifies byte-identical validation pixels across dataset rebuilds.
- **Optuna pre-tuning:** Six (model, dataset) pairs were tuned at L3 Moderate before the 186-cell campaign launched; the winning hyperparameters were frozen for every Phase A/B/C cell of that pair. Winners under `artifacts/best_hparams/{model}_{dataset}.json`.
- **Engine:** PyTorch Lightning in `src/lightning/` (THzClassifier, THzDataModule). `bf16-mixed` on Blackwell (sm_120) RTX 5070; `32-true` on CPU.
- **Weight privacy:** Trained checkpoints (`*.ckpt`), the Optuna SQLite store, and dashboard thumbnail caches are local-only, gitignored, and claudeignored. Analysis tooling reads `metrics.json` / `metrics.csv` / the tracker artifacts only — never the binary weights.

Appendix A — Full 186-cell Result Table

See `Final_Exp.md` (auto-regenerated from `runs/final/` by `scripts/update_final_exp.py`) for the canonical 186-cell tracker with PSNR / SSIM / Started / Duration columns. This report's tables summarize the same data with focus on `val_acc`.

Appendix B — Story-Driven Methodology Timeline

The campaign executed under a 9-story execution roadmap (US-006 through US-014), each closing a (phase, model) sub-campaign with operator-gate HALTs between stories. After US-014 closure, US-015 (end-of-campaign verification + three phase-boundary SYNCHRONIZER pushes + SECURITY leak audit) completes the campaign.

Closure decisions were recorded in `progress.txt` as `Iteration N` blocks (currently 25 iterations across Phase B and Phase C dispatches), each with the dispatch command, background task id, first-pass readings, pathology-guard outcomes, and operator decisions. The §6.4 retry decline streak (5-for- across all Phase B/C CNN dispatches + the one TransNeXt Phase C dispatch with sentinels) is documented inline.

Appendix C — Repository Pointers

- `Final_Exp.md` — 186-cell tracker (auto-regenerated)
- `artifacts/Final_Exp.json` — JSON-shaped tracker (dashboard data source)
- `artifacts/Final_Exp.html` — interactive dashboard with sortable rows + filters

- docs/phase_a.md / docs/phase_c.md — LIBRARIAN per-phase docs
- progress.txt — full iteration log (Iterations 1-25)
- PRD.md — 9-story execution roadmap (US-006 through US-015)
- CLAUDE.md — protocol invariants + multi-agent guidance
- scripts/build_final_exp_report.py — this report generator
- scripts/render_final_exp_pdf.py — Markdown → PDF renderer

Report generated 2026-05-20 from artifacts/Final_Exp.json (state of disk after Iteration 25 closure). Regenerate with `python scripts/build_final_exp_report.py`.